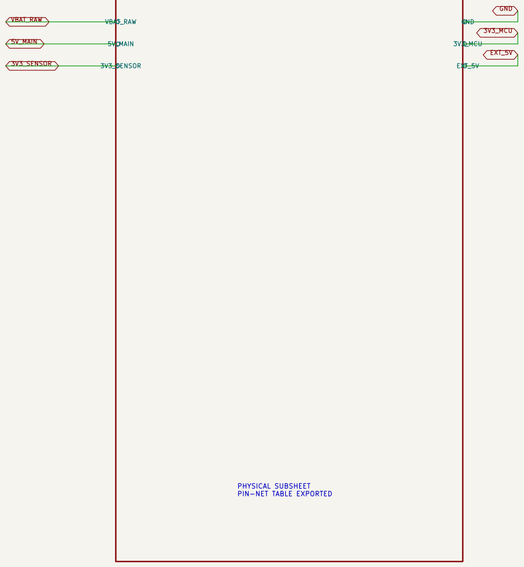
X100 CLEAN-SHEET | PHYSICAL SCHEMATIC

VBAT 3S-6S -> 5V_MAIN -> 3V3_MCU / 3V3_SENSOR / protected EXT_5V

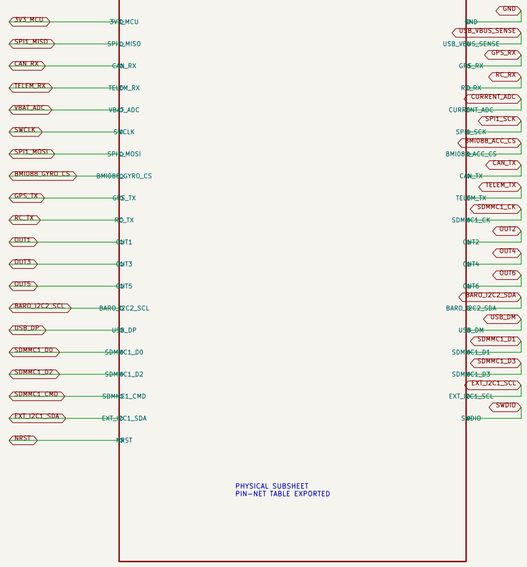
USB: DATA ONLY + VBUS SENSE. USB_VBUS never joins 5V_MAIN.

Physical pins are wired in subsheets. Only nonblocking board/test qualifications are listed in GATE1 exports.

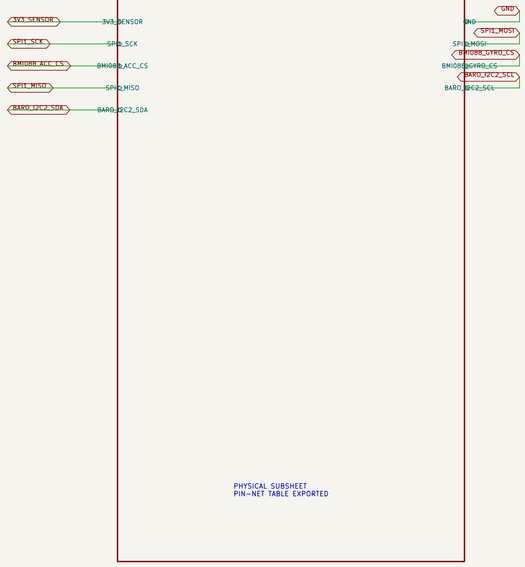
01_POWER



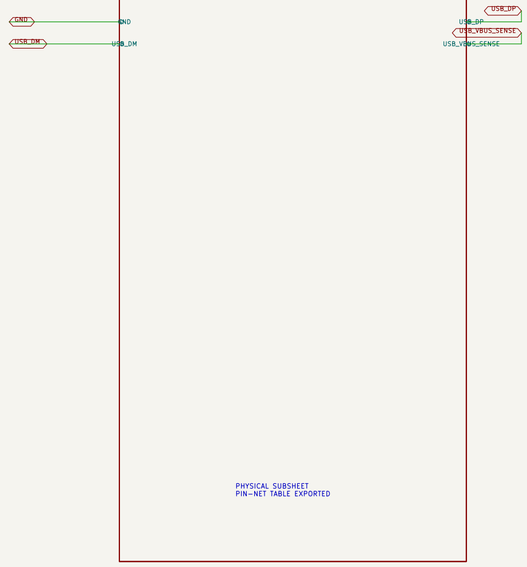
02_MCU



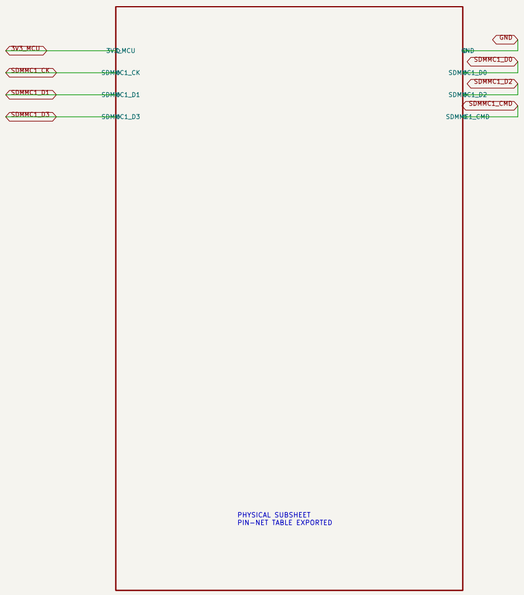
03_SENSORS



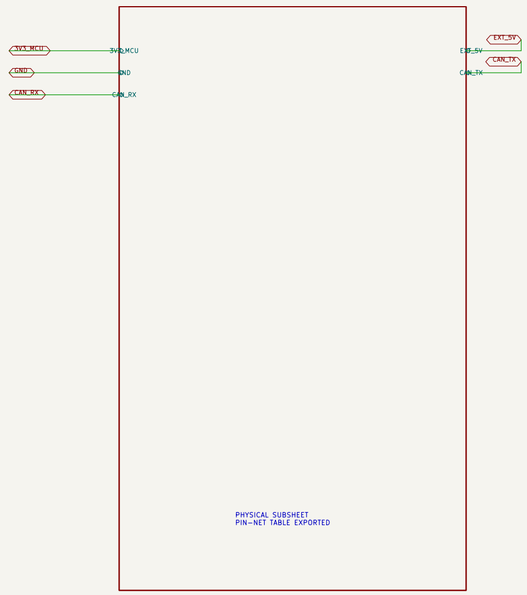
04_USB



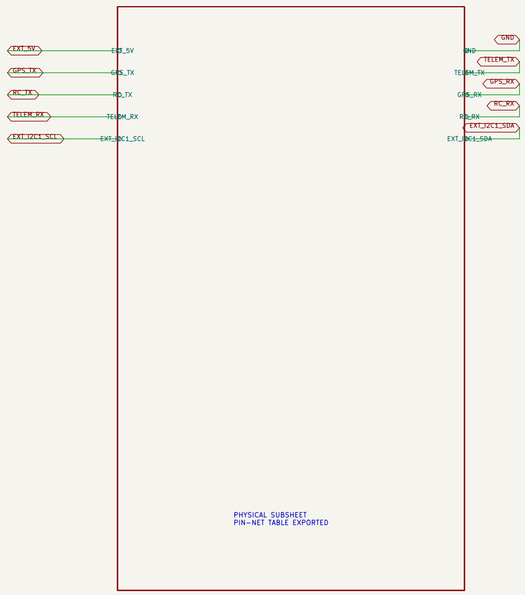
05_SD



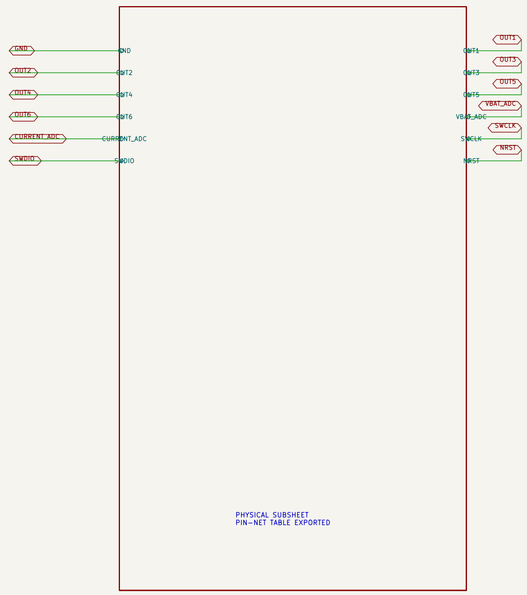
06_CAN



07_UART_IO



08_PWM_ADC_DEBUG



PHYSICAL SCHEMATIC; PCB HOLD; see local unresolved inputs

PHYSICAL SCHEMATIC; PCB HOLD; see local unresolved inputs

X100 Clean-Sheet

Sheet: /
File: X100_CleanSheet.kicad_sch

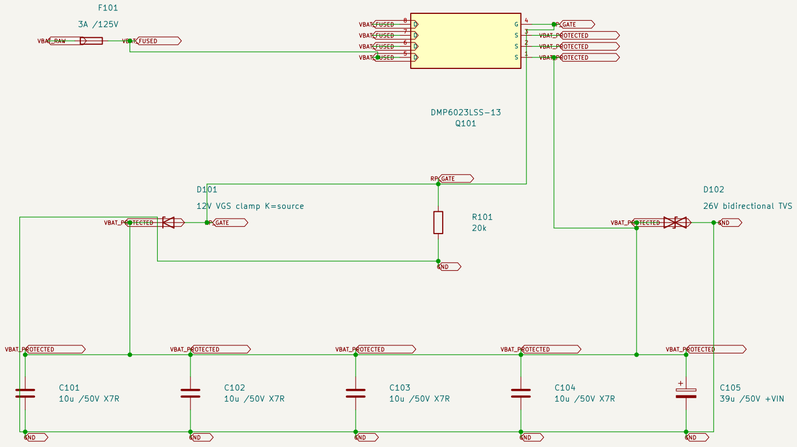
Title: X100 Clean-Sheet / hierarchy

Size: A2	Date: 2026-10-04
KiCad E.D.A. 10.0.6	**Rev: Physical-1**
	Id: 1/9

01 电源管理 / POWER MANAGEMENT

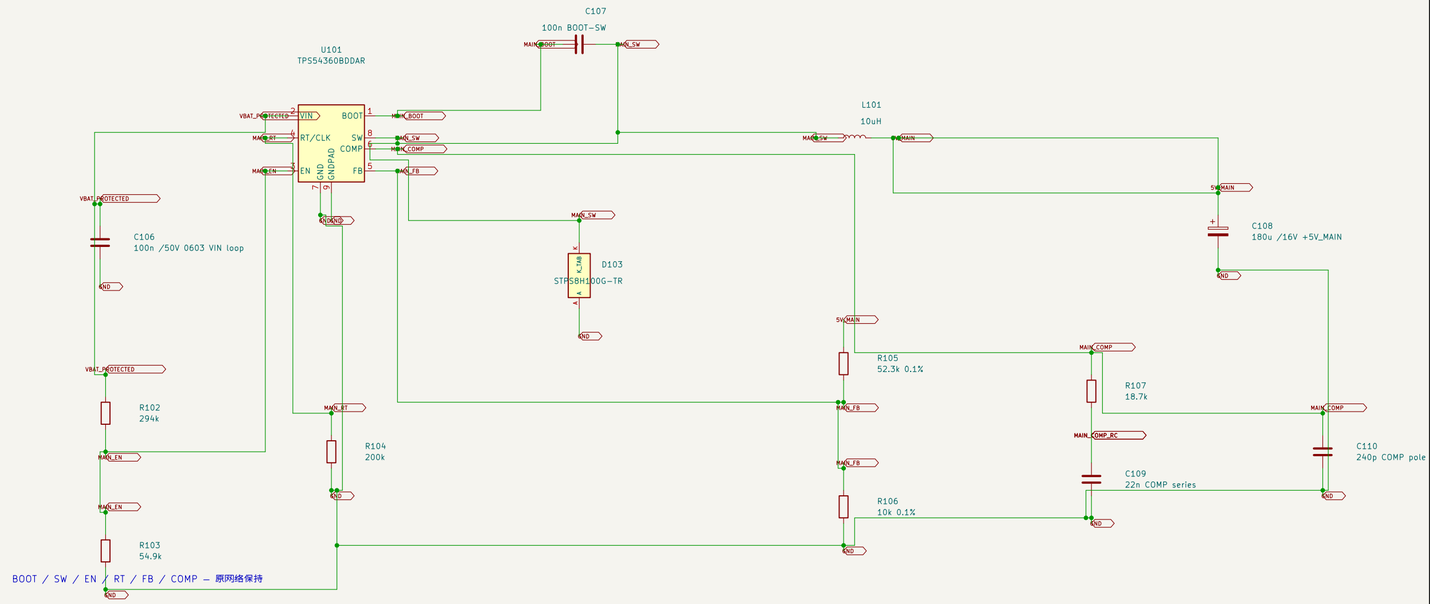
MAIN 5V: battery/bench only, 3S–6S. USB is never a supply.

VBAT 输入与保护 / VBAT INPUT AND PROTECTION

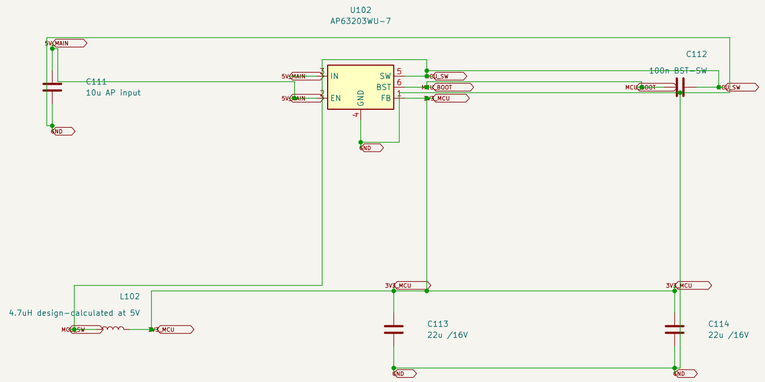


VBAT_RAW → VBAT_FUSED → VBAT_PROTECTED

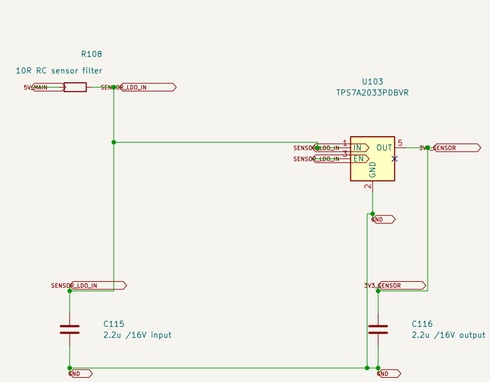
主 5V 降压电源 / MAIN 5V BUCK



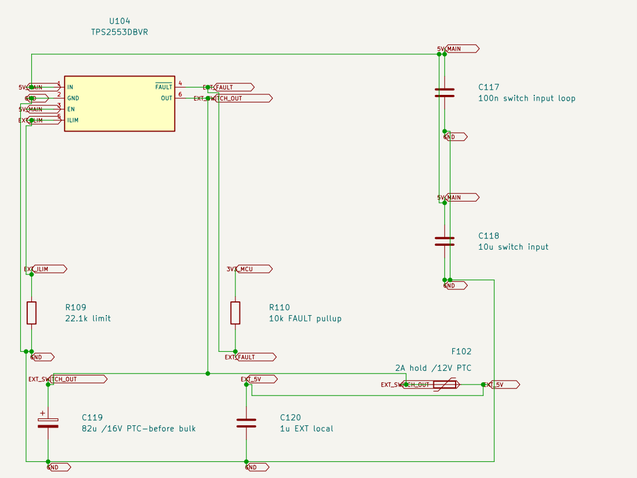
MCU 3.3V 电源 / MCU 3V3 SUPPLY



传感器 3.3V 电源 / SENSOR 3V3



受控外部 5V / CONTROLLED EXTERNAL 5V



VBAT / MAIN 5V



MCU / SENSOR POWER



EXTERNAL 5V / GND



TP101 VBAT_RAW
TP102 VBAT_FUSED
TP103 VBAT_PROTECTED
TP104 5V_MAIN
TP105 3V3_MCU
TP106 SENSOR_LDO_IN
TP107 3V3_SENSOR
TP108 EXT_SWITCH_OUT
TP109 EXT_5V
TP110 EXT_FAULT
TP111 GND

#FLG101 PWR_FLAG
#FLG102 PWR_FLAG
#FLG103 PWR_FLAG
#FLG104 PWR_FLAG
#FLG105 PWR_FLAG
#FLG106 PWR_FLAG
#FLG107 PWR_FLAG
#FLG108 PWR_FLAG

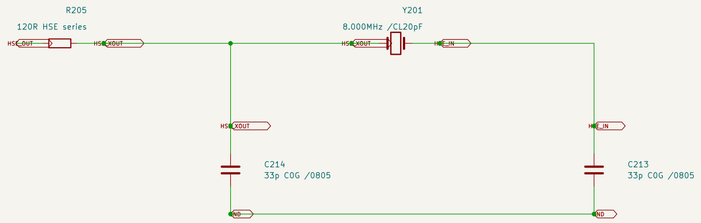
TPS2553 constant-current limiting; recovers after overload removal, subject to thermal protection.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic
X100 Clean-Sheet

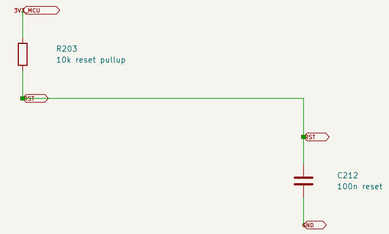
Sheet: /01_POWER/	
File: 01_POWER.kicad_sch	
Title: 01_POWER	
Size: A2	Date: 2026-10-04
KiCad E.D.A. 10.0.6	**Rev: Physical-1**
	Id: 2/9

02 主控与时钟 / MCU AND CLOCK

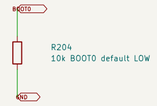
HSE 晶振 / HSE CLOCK · 8.000 MHz



复位 / RESET



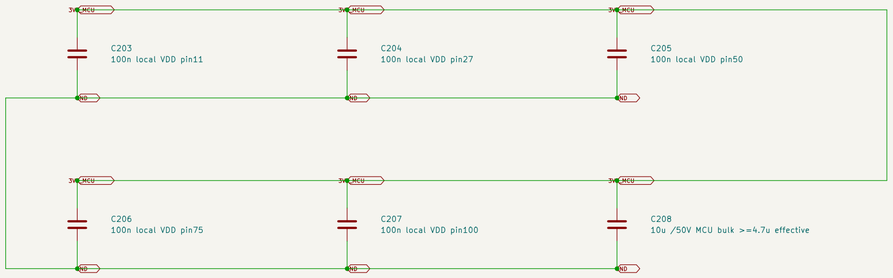
启动配置 / BOOT CONFIGURATION



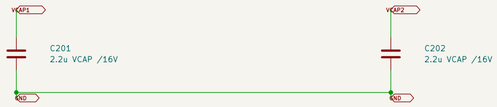
BOOT0 默认LOW · 恢复方式见页底说明

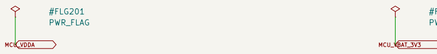
STM32H743 主控与去耦 / MCU CORE AND DECOUPLING

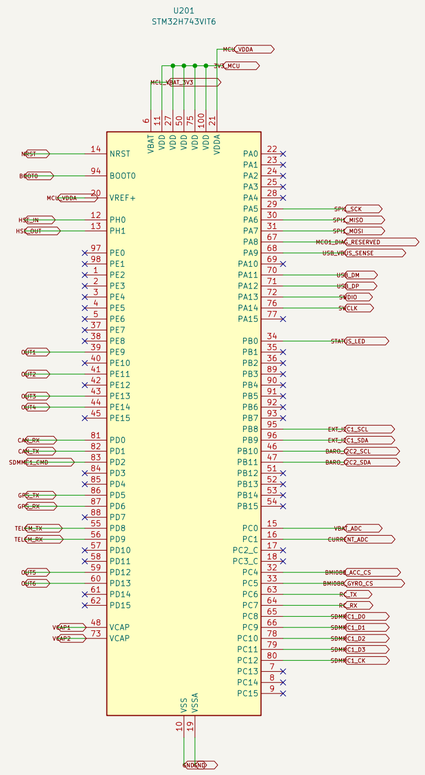
VDD 去耦 / VDD DECOUPLING



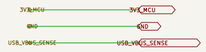
VCAP 核心稳压 / CORE REGULATOR



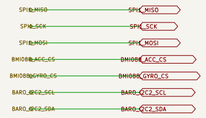




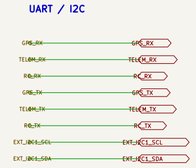
POWER / CONTROL



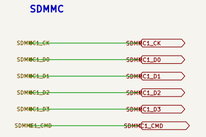
SPI / IMU / BARO



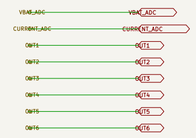
UART / I2C



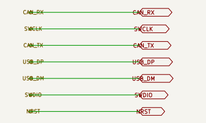
SDMMC



PWM / ADC

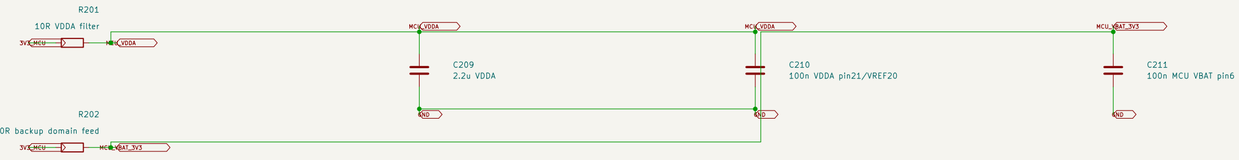


USB / CAN / DEBUG





模拟与备份供电 / ANALOG AND BACKUP POWER



状态与电源指示 / STATUS AND PWR LED



STM32H743VIT6 LQFP100. Frozen B allocation; unused GPIO has explicit NC.

BOOT0 recovery: strap TP_BOOT0 to TP_3V3 with power off; then reset. No flight jumper.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet

Sheet: /02_MCU/
File: 02_MCU.kicad_sch

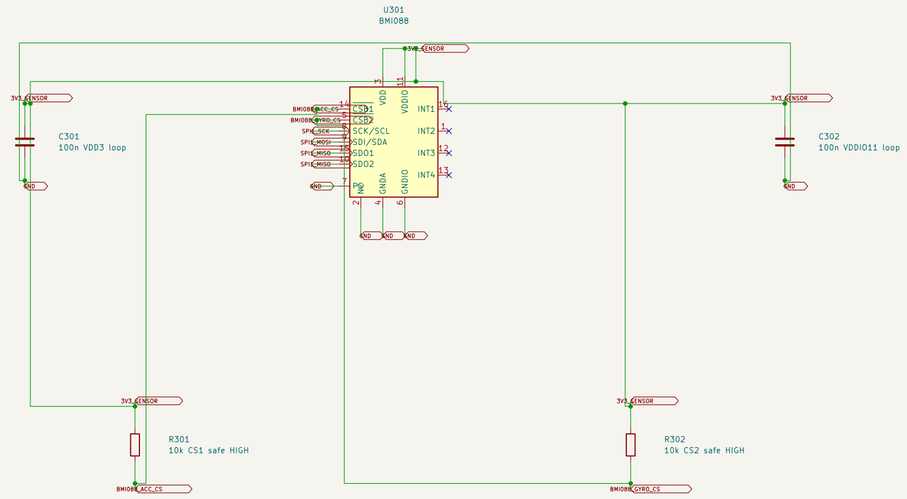
Title: 02_MCU

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 3/9

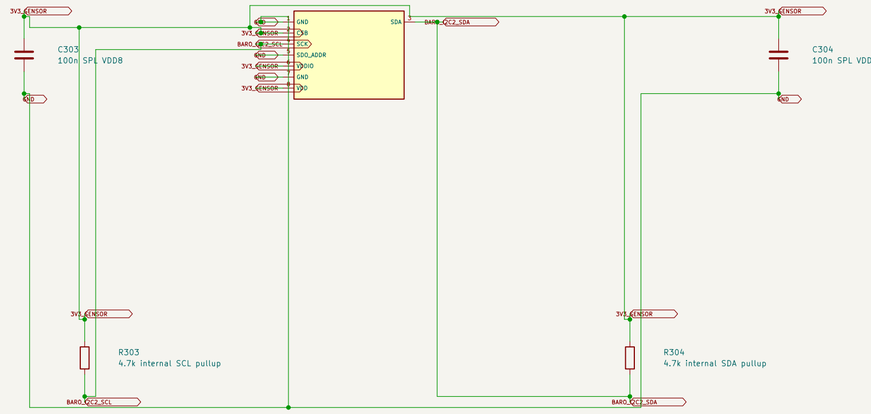
03 传感器 / SENSORS

Single BMI088 SPI1 and single SPL06-001 I2C2. No compass/second IMU.

BMI088 惯性测量 / BMI088 IMU · SPI1

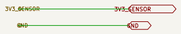


SPL06-001 气压计 / BAROMETER · I2C2

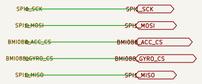


传感器接口与测点 / SENSOR INTERFACES AND TEST

POWER / GND

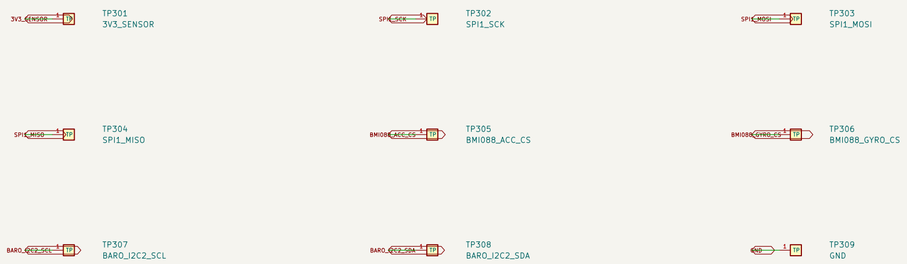


SPI1 / IMU



I2C2 / BARO





SPL: CSB2=VDDIO; SDO5=GND => I2C0x76. BMI: PS7=GND; NC2=GND.

SPI initialization must toggle accel CS and discard dummy read. Sensor-off GPIO must not drive.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /03_SENSORS/
File: 03_SENSORS.kicad_sch

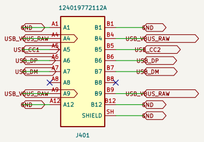
Title: 03_SENSORS

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 4/9

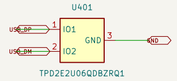
04 USB通信与保护 / USB INTERFACE AND PROTECTION

USB DATA ONLY + VBUS SENSE. Battery/bench power required.

USB Type-C 接口 / USB TYPE-C CONNECTOR



USB 数据线 ESD / DP-DM ESD



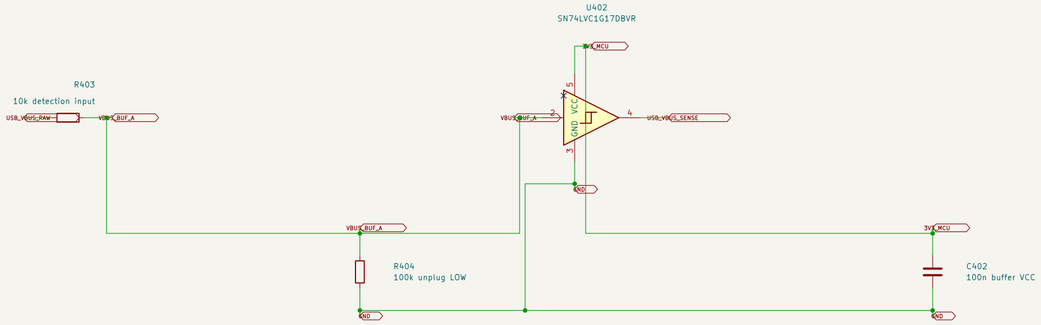
VBUS 保护 / USB VBUS PROTECTION



CC 配置与保护 / CC CONFIGURATION AND ESD



VBUS 检测 / VBUS SENSE



USB DP / DM



VBUS / POWER / GND





TP401
USB_VBUS_RAW



TP402
USB_VBUS_SENSE



TP403
USB_DP



TP404
USB_DM



TP405
GND

PA9 is software VBUS sense: configure GPIO input/session control; no raw5V on an unpowered MCU.

NO power OR, NO USB supply diode, NO USB connection to any main/sensor rail.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /04_USB/
File: 04_USB.kicad_sch

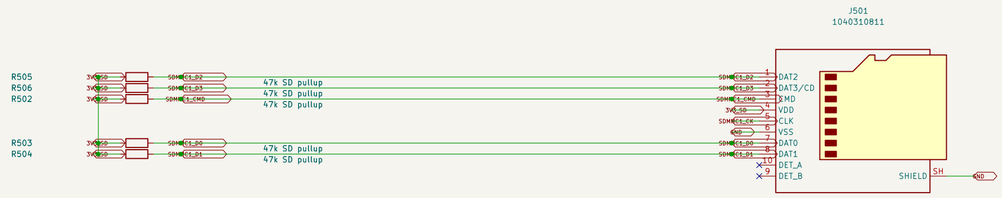
Title: 04_USB

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 5/9

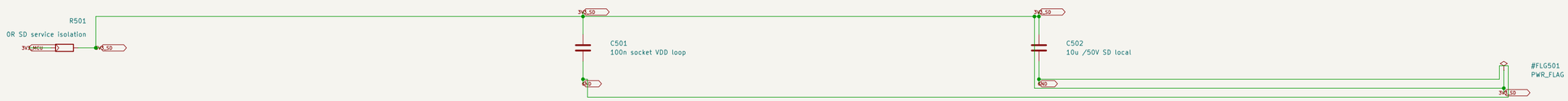
05 microSD数据存储 / MICROSD STORAGE

SDMMC1 4-bit, 3.3V only. Contact9/10 detect NC; no detect GPIO.

microSD 接口与信号上拉 / MICROSD AND SDMMC PULLUPS



本地供电与去耦 / LOCAL POWER AND DECOUPLING



STM32 SDMMC1 ↔ microSD · 3.3V / 4-bit

SDMMC / SIGNALS

POWER / GND

TP501 3V3_SD
TP502 SDMMC1_CK
TP503 SDMMC1_CMD
TP504 SDMMC1_D0
TP505 SDMMC1_D1
TP506 SDMMC1_D2
TP507 SDMMC1_D3
TP508 GND

No arbitrary damping resistor or 1.8V UHS. Series damping only after signal-integrity evidence.

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet

Sheet: /05_SD/
File: 05_SD.kicad_sch

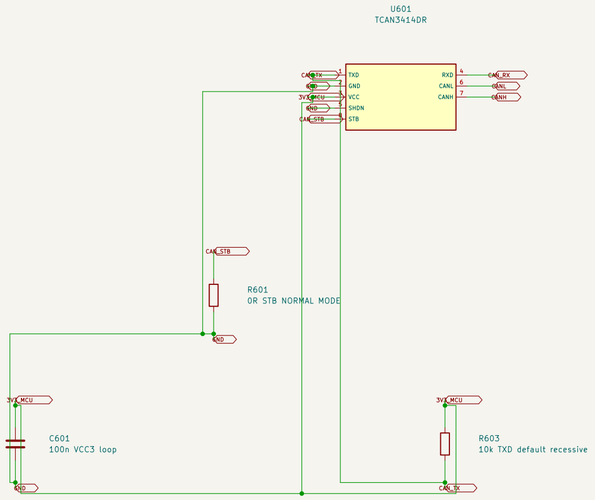
Title: 05_SD

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 6/9

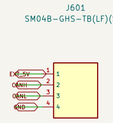
06 CAN通信 / CAN INTERFACE

TCAN3414DR 3.3V: Classical CAN only. SHDN=0,STB=0: Normal mode.

CAN 收发器 / CAN TRANSCEIVER



CAN 总线接口 / CAN BUS CONNECTOR



120Ω 终端 / CAN TERMINATION · 默认OPEN



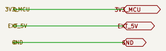
CANH → R602 → CAN_TERM → JP601 → CANL

按口与测点 / INTERFACES AND TEST POINTS

MCU / CONTROL



CANH / CANL / POWER





TP601
CAN_TX



TP602
CAN_RX



TP603
CANH



TP604
CANL



TP605
GND

JP601 CAN TERM / 120R: manufacture OPEN (disabled). Manually solder bridge only at bus end (enabled). NO PASTE; wick clean to restore OPEN.

Built-in bus standoff +/-58V: receiver common mode +/-30V; bus-pin IEC ESD +/-10kV.

EXTERNAL_CAN_TVS=NOT_REQUIRED_FOR_V1. Whole-board IEC qualification deferred to test.

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet

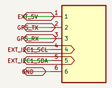
Sheet: /06_CAN/		
File: 06_CAN.kicad_sch		
Title: 06_CAN		
Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 7/9

07 串口与通用IO / UART AND GENERAL I/O

External5V is protected EXT_5V; all UART/I2C signals3.3V.

GPS 接口 / GPS INTERFACE

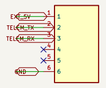
J701
SM06B-GHS-TB(LF)(SN)



GPS: FC_TX to receiverRX

遥测接口 / TELEMETRY INTERFACE

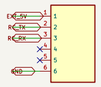
J702
SM06B-GHS-TB(LF)(SN)



TELEM:4/5 NC no flow control

RC 接收机 / RC RECEIVER · CRSF

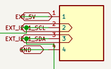
J703
SM06B-GHS-TB(LF)(SN)

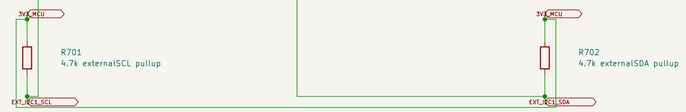


RC: CRSF3.3V logic:4/5 NC

外部 I2C / EXTERNAL I2C

J704
SM04B-GHS-TB(LF)(SN)

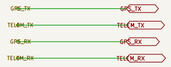




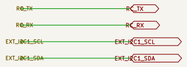
externalSCL pullup

externalSDA pullup

GPS / TELEM

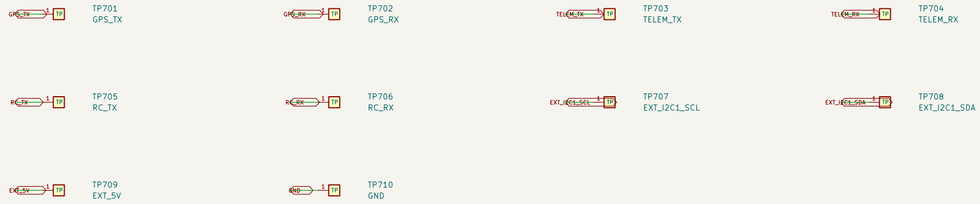


RC / EXTERNAL I2C



POWER / GND





GPS and externalI2C share I2C1. Combined pullups/cable capacitance must satisfy rise time.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet
Sheet: /07_UART_IO/
File: 07_UART_IO.kicad_sch

Title: 07_UART_IO

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 8/9

08 PWM / ADC / 调试 · PWM / ADC / DEBUG

PWM signal+GND only. Custom SWD5pin. Pack current NEVER passes through board.

六路 PWM 输出 / SIX PWM OUTPUTS

J801
OUT1 /GND



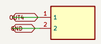
J802
OUT2 /GND



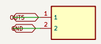
J803
OUT3 /GND



J804
OUT4 /GND



J805
OUT5 /GND

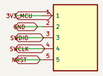


J806
OUT6 /GND



SWD 调试与电池输入 / SWD AND BATTERY INPUT

J807
SWD:1REF2GND3IO4CLK5NRST

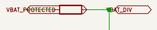


J808
BATTERY/bench3S–6S ONLY



电压采样 / VOLTAGE SENSE · ADC PC0

R801
150k 0.1%



R803
1k ADC series



VBAT_DIV

R802
10k 0.1% scale16

GND

VBAT_ADC_PRE_PROTECTION

C801
100n ADC LPF

GND

电流输入采样 / CURRENT INPUT · ADC PC1

J809
CURRENT monitor only



R805
221k 1% >=100V

CURRENT_DIV

R806
20k 1% scale20/42.1

GND

R807
1k current limit

CURRENT_ADC_PRE_PROTECTION

C802
100n X7R 50V

GND

掉电隔离与保持 / POWERED–OFF ISOLATION

U801
TMUX1511PWR

3V3_MCU

C805
100n U801 VDD14

GND

3V3_MCU

R811
1k rail discharge 3.3mA

GND

ADC 输出滤波 / ADC OUTPUT FILTERS

R804
1k isolate ADC reservoir

VBAT_ADC

C803
22n X7R 50V ADC reservoir

GND

VBAT_ADC

R809
1.3M 1% discharge/IOFF bound

GND

R808
1k isolate ADC reservoir

1.3M 1% discharge/IOFF bound

CURRENT_ADC

C804
22n X7R 50V ADC reservoir

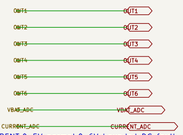
GND

CURRENT_ADC

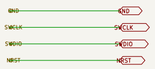
R810

GND

PWM / ADC



DEBUG / POWER



CURRENT:0..5V normal,0..6V bounded DC fault, shared GND; no VBAT/negative-fault rating.

PC0/PC1 are FT_a. No rail-clamp diode added: external U801 powered-off isolation + bleed.

150k/10k remains nominal1/16; finite1.3M loading gives actual scale16.1401. Calibrate.

TP801
VBAT_DIV

TP802
VBAT_ADC_PRE_PROTECTION

TP803
VBAT_ADC

TP804
CURRENT_IN_RAW

TP805
CURRENT_ADC

TP806
SWDIO

TP807
SWCLK

TP808
NRST

TP809
GND

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /08_PWM_ADC_DEBUG/
File: 08_PWM_ADC_DEBUG.kicad_sch

Title: 08_PWM_ADC_DEBUG

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 9/9